

compiled later in his career but show subjects of earlier interest—anatomy became one of his primary concerns. It is likely that da Vinci began conducting his own dissections after 1483, when he entered the employ of Duke Ludovico Sforza in Milan, which was a leading center of medical investigation. He later attested to having performed some 30 dissections during his career.

Knowing how to see

Da Vinci believed that painting itself was a science, primarily because of art's capability of being, in the right hands, "the sole imitator of all the manifest works of nature." Observation was the essential tool of this discipline, which da Vinci summed up in his maxim *saper vedere*—which means "knowing how to see" one's subject. Unlike sculpting, drawing and painting require the artist to render a three-dimensional scene or object in two dimensions,

"He who **loves practice without theory** is like a sailor **with no rudder.**"

Leonardo da Vinci

meaning that artists of these media must achieve mastery of mathematical and optical principles.

A devotee of the mathematical basis of his art, da Vinci believed that geometry and proportion were essential to achieve perspective. He was particularly interested in

LEONARDO FIBONACCI

Italian mathematician
Leonardo Fibonacci
popularized Hindu-Arabic numbers.



Born in Pisa, Fibonacci (c.1170–1250) learned the Hindu-Arabic numeral system while helping his merchant father in Algeria. He then studied with Arab mathematicians, writing up his findings in *Liber Abaci* and *Practica Geometriae*. He also devised the Fibonacci sequence, in which each number is the sum of the two preceding it: 0, 1, 1, 2, 3, 5, 8, and so on. Squares of each value can be drawn and aligned to create a spiral.

